

Response of Geomagnetic Field at Indian Antarctic Station Maitri, to Increasing Electromagnetic Disturbance in Geospace

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Introduction

The Indian Antarctic station Maitri has the geographic coordinates $70^{\circ}46'S$ and $11^{\circ}45'E$, with corrected geomagnetic coordinates as $66^{\circ}84'S$, $56^{\circ}29'E$ (according to IGRF, 1990). Maitri lies in a sub-auroral position during magnetically quiet times, *i.e.*, it lies just equatorward of the oval-shaped region in the southern hemisphere which experiences frequent auroral displays. The geomagnetic signatures seen at Maitri at such times are predominantly those of the Solar Quiet (Sq) currents which flow at low and mid-latitudes, and only marginally those of the Auroral Electrojet (AE) currents. Both the Sq and AE types of currents flow in the Earth's ionosphere at an altitude of 100-110 km.

During times of electromagnetic disturbance in geospace Maitri comes under the auroral oval, and then it experiences the geomagnetic signatures of (i) the auroral electrojet current system mentioned above and (ii) the field-aligned currents which link the magnetosphere to the ionosphere. The position of Maitri with respect to the auroral oval during magnetically Quiet and Disturbed times is shown in Fig. 1(a) and 1(b) respectively. These figures represent the southern hemisphere as viewed from above the south magnetic pole. The oval-shaped auroral region surrounding the south pole generally lies between $65^{\circ}S$ and $70^{\circ}S$ geomagnetic latitude during Quiet conditions, and shifts to latitudes roughly between $60^{\circ}S$ and $65^{\circ}S$ at disturbed times.

The current systems which operate within this auroral region are shown in Fig. 2. Also marked on the figure are the local times referring to midnight (00 LT), noon (12 LT), dawn (06 LT) and dusk (18 LT) sectors. With reference to the north-south orientation of the Earth's axis, the auroral electrojet current in the dawn sector flows in the westward direction while in the dusk sector it flows in the eastward direction. These dawn and dusk AE currents are shown by thick

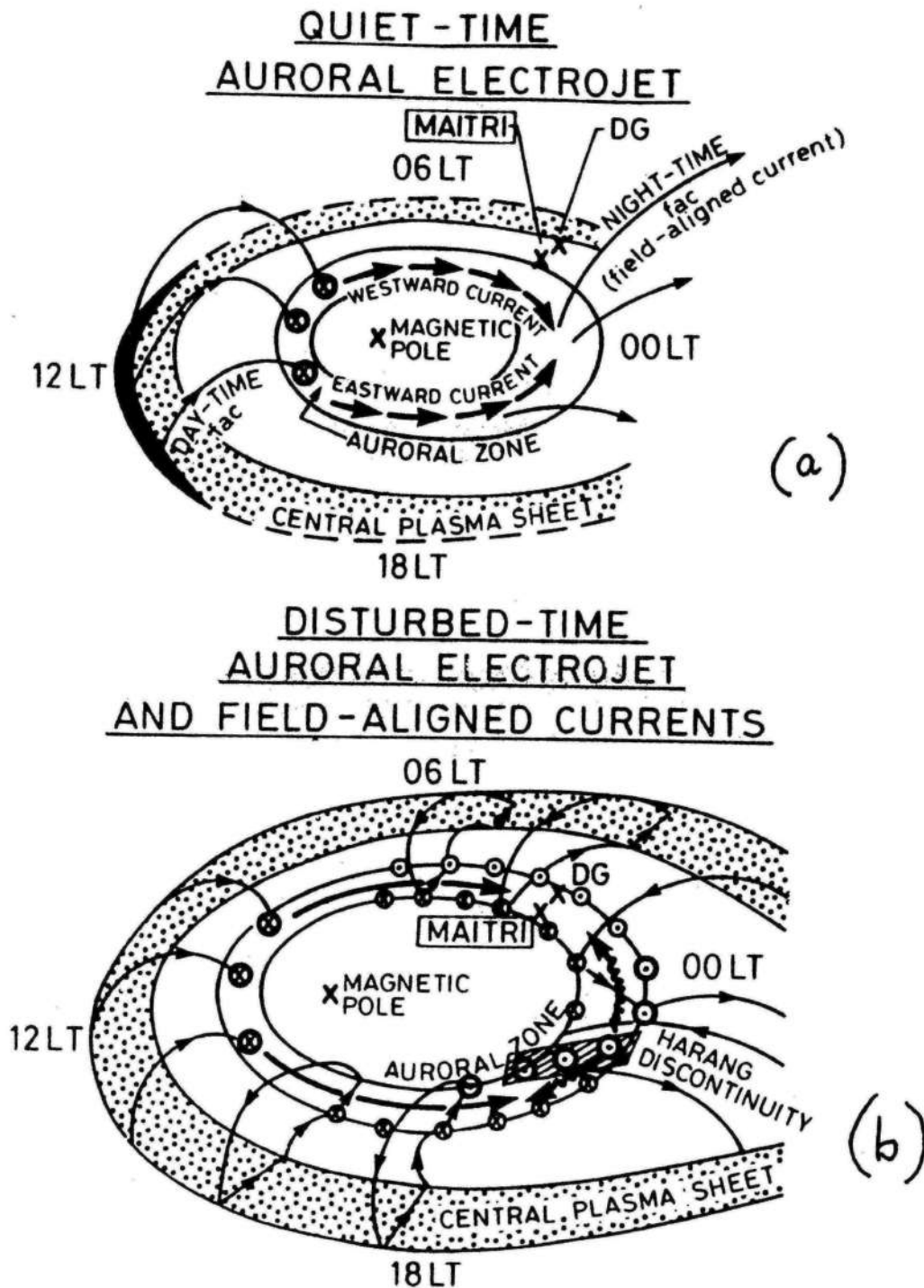


Fig. 2. The current systems which operate within the auroral region are the westward and eastward auroral electrojets in the ionosphere and the Field Aligned Currents (FAC) from the Central Plasma Sheet (CPS). (a) During Quiet times FAC flow into the auroral ionosphere from the CPS, mainly in the noon and midnight time sectors. (b) During Disturbed time the auroral oval expands, FAC intensifies and expands into the dawn and dusk time sectors

Observations

For the present study we have selected three days from each month of 1992, representing Quiet (Q), Moderately Disturbed (MD) and Disturbed (D) magnetic conditions. The X, Y and Z components measured on these days are shown in Figs. 3 to 7. Each figure consists of 6 blocks; the upper three blocks representing the X, Y and Z variations for one month and the lower three representing these for the next month. The quiet day variations are shown by full lines, the moderately disturbed days by dotted lines, and the disturbed variations by dashed lines. Fig. 3 shows these variations for the Equinoctial months of March and April 1992. Figs. 4 to 7, similarly depict observations for May 1992 to December 1992.

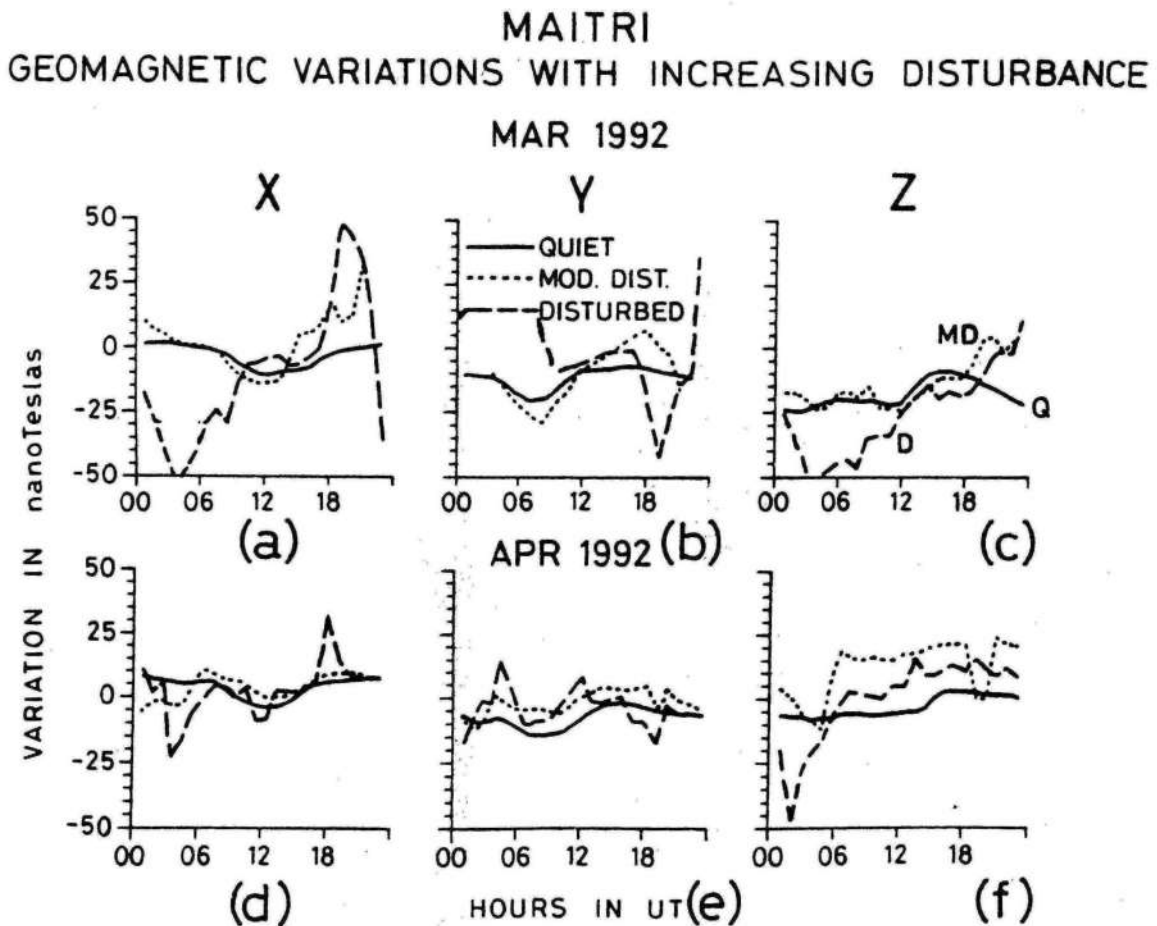


Fig. 3. Hourly variations of the X, Y and Z geomagnetic components on a Quiet (full line), a Moderately Disturbed (dotted line) and a Disturbed (dashed line) days for the equinox months March and April 1992. Note in the X-component a clear decrease in the dawn hours and a clear increase in the dusk hours which signals the entrance of Maitri into the auroral oval. This feature is seen for the other months too in the figures which follow

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MAY 1992

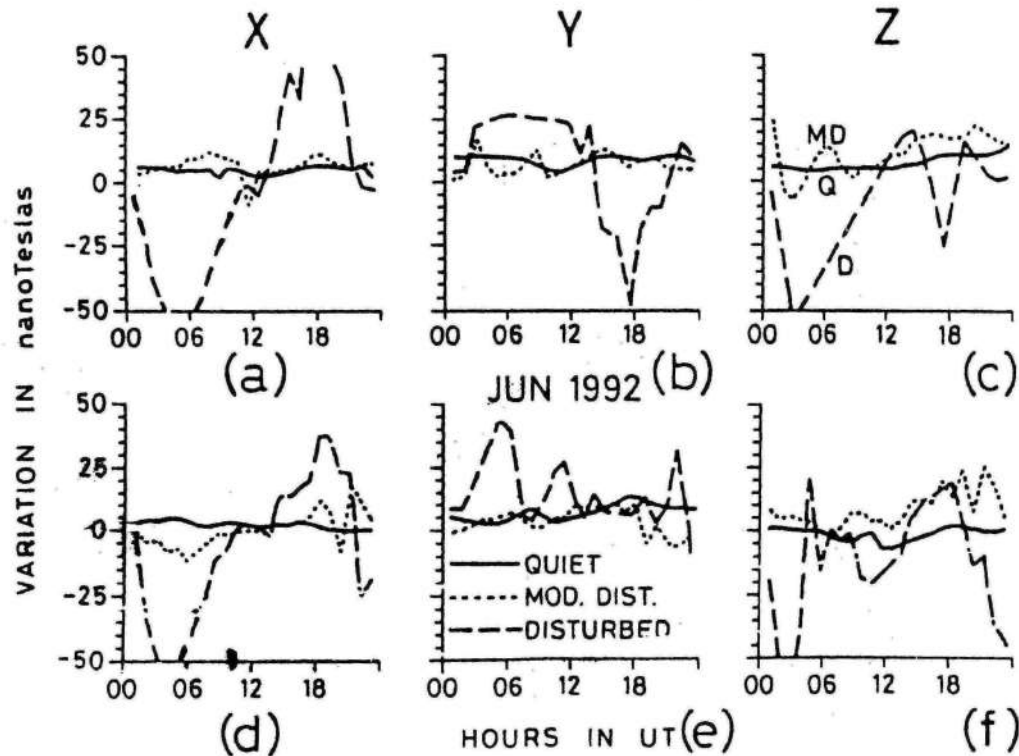


Fig. 4. Hourly variations of X, Y and Z for the Antarctic Winter months May and June 1992. Notice how the flat variation of Y on Quiet days is replaced by large variations on Disturbed days showing presence of strong FAC over Maitri

Variations of the X-Component with Magnetic Activity

Fig. 3(a) shows that the Quiet day (Q) variations of the X component show negative values (southward component) during noon hours. Between 18 and 06 UT, the variations remain much in the vicinity of the zero line suggesting very weak current flow. The Q curves for the X-component in Fig. 3, thus, appear to be the signature of the Sq current system which at these high latitudes has a westward direction in the daytime and hence gives a southward deflection in the noon hours. Since the Sq current is very weak in the night hours and is directed eastward at these latitudes the magnetic signature on ground will be seen as a mild positive variation. This is what is seen at Maitri. The Disturbed day curve for the X-component differs greatly from the Q day curves in that (i) the flat trend of prenoon hours is replaced by a large negative (southward) variation in the X-component and (ii) the flat variation of the Q-day curve in night time hours is replaced by a large positive variation in the X-component. These features suggest that Maitri has now shifted into the auroral oval and sees

the signatures of intense Westward Auroral Electrojet (WAE) in the dawn hours and intense Eastward Auroral Electrojet (EAE) in the dusk hours. The geomagnetic signatures show that Maitri experiences the effect of the Sq current system during Q times and that of the auroral electrojet during D times. During MD times the X-variations in Fig. 3(a) understandably show variations which lie between Q and D conditions.

The features discussed for Fig. 3(a) as regards changes in the X-component as a function of Q, MD and D days are also reflected in the X-component variations for April 1992 (Fig. 3d); for May 1992 (Fig. 4a); for June 1992 (Fig. 4d); for July 1992 (Fig. 5a); for August 1992 (Fig. 5d); for September 1992 (Fig. 6a where again forenoon values for D days are missing); for October 1992 (Fig. 6d); for November 1992 (Fig. 7a) and for December 1992 (Fig. 7d).

It is seen that the nature of the Q day variations for the X-component changes with season, *i.e.*, the Antarctic winter months of May, June, July and August show almost flat variations in the vicinity of zero, while during Antarctic summer months (November and December) large noon time negative variations in X-component are seen.

Irrespective of season, however, the large negative variations in the prenoon hours, and the large positive variations in the afternoon hours stand out during D days, clearly indicating that Maitri, during D days in all seasons, experiences the signatures of dawnside westward auroral electrojet currents and duskside eastward auroral electrojet currents.

Variations of the Y-Component with Magnetic Activity

The Q time variations of the Y-component during March and April 1992 are shown in Fig. 3(b) and 3(e) as full lines. These show a pronounced negative in the pre-noon hours, but this levels off in the afternoon hours. With the onset of disturbance variations set in and the disturbed thick lines show a tendency for a positive (eastward) peak in the pre-dawn hours and a negative (westward) dip in the dusk hours. This trend is seen in Figs. 4(b) and 4(e) for May and June 1992; for August 1992 in Fig. 5(e); for September and October 1992 in Figs. 6(b) and 6(e); and for November and December 1992 in Figs. 7(b) and 7(e). July 1992 (Fig. 5b) is an exception in that the positive occurs in the pre-noon hours rather than at pre-dawn hours. This, however, may be a characteristic of that particular day.

The changes in the Y-component with the onset of magnetic disturbance suggest that

- (i) the Y-component variations are substantially increased by magnetic disturbance;

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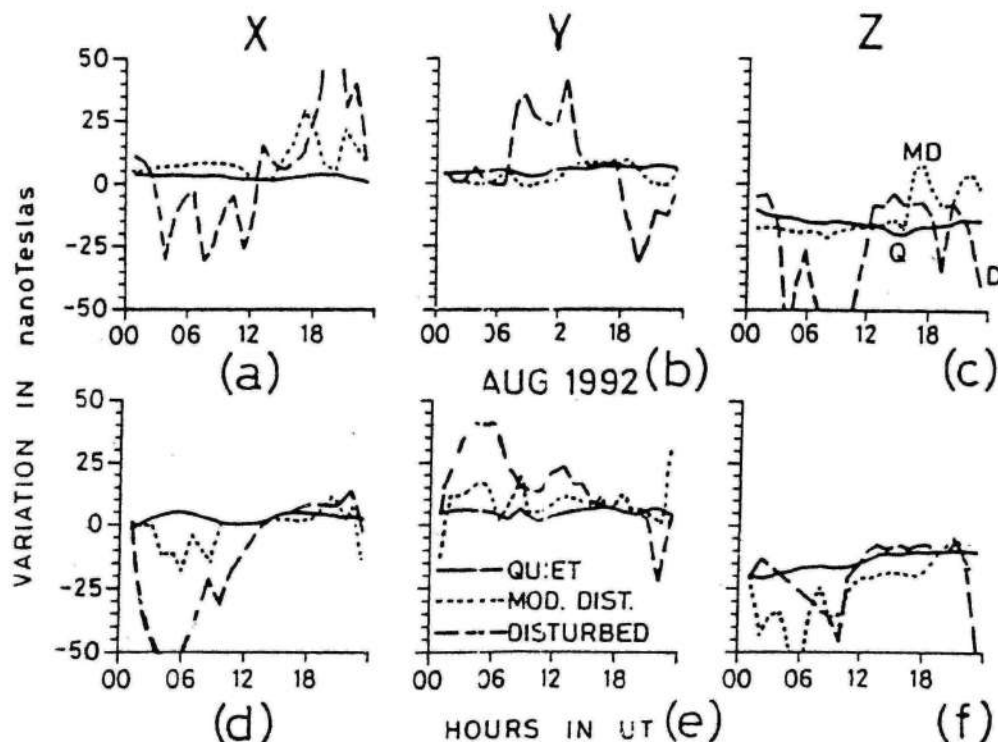


Fig. 5. Hourly variations of X, Y and Z for the Antarctic Winter months July and August 1992

- (ii) the large variations in Y-component which occur in the pre-dawn and dusk hours could be due to field-aligned currents (FAC) which flow into the ionosphere from the outer magnetosphere. Such FAC are known to intensify and spread to lower latitudes during disturbed conditions. Maitri evidently comes under their influence at such times; and
- (iii) from the variations in the Y-component alone it is not possible to judge the direction of the FAC. The FAC could be flowing in from the magnetosphere or it could be flowing out from the ionosphere – the direction of this FAC over Maitri could decide whether a positive or a negative variation is seen in the Y-component. The nature of the Y-variation is also decided by the location of Maitri to the north or the south of this FAC.

A marked seasonal variation is also seen in the Q day variations of the Y-component. The Q day variations are rather flat for the Antarctic winter months of May, June, July, August, and even September (Figs. 4b, 4e, 5b, 5e and 6b, respectively). As summer sets in, the variation becomes pronounced,

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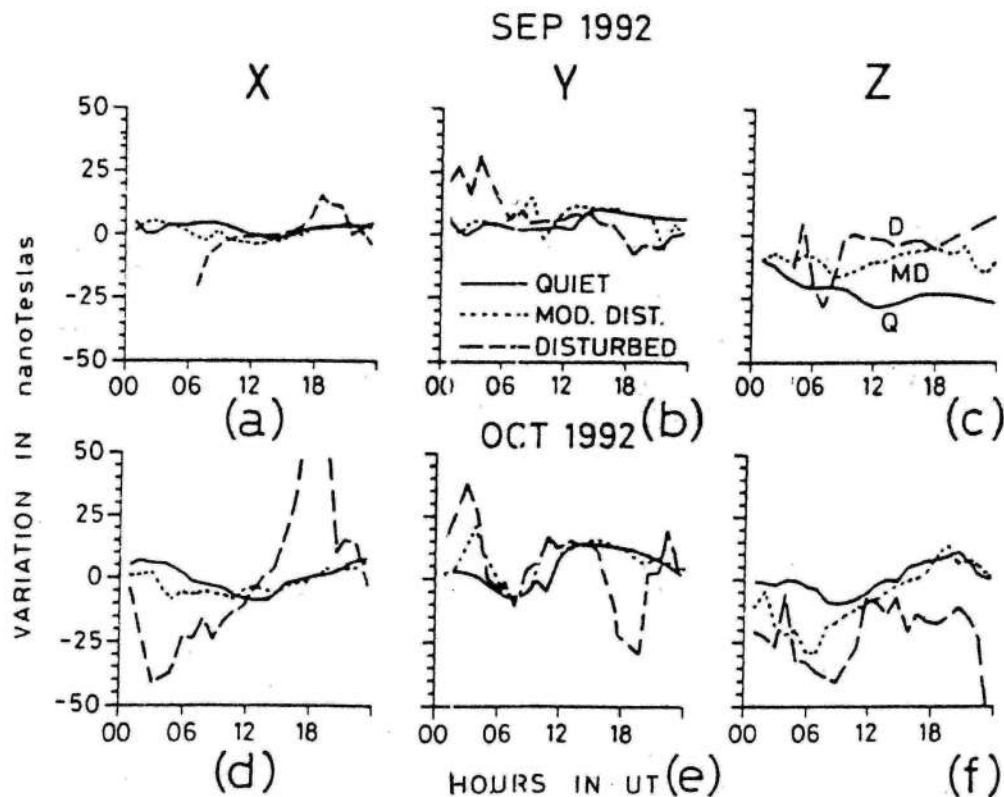


Fig. 6. Hourly variations of X, Y and Z for the equinoctial months September and October 1992. Note how the small variation of Z on Quiet days is replaced on Disturbed days by large variations in the pre-noon and post-noon hours

with a negative in the pre-noon hours and a positive in the post-noon hours. This is seen for October, November, December 1992, and even for March, April 1992 (Figs. 6e, 7b, 7e, 3b and 3e, respectively).

This could be the signature of the meridional component of the Q time Sq current system referred to earlier. This meridional Sq current is directed northwards in the pre-noon sector and hence would cause negative (westward) change in the geomagnetic Y-component (Vestine, 1960). It is directed southwards in the post-noon sector and then it causes positive (eastward) change in the Y-component. These are the signatures which are seen in the Y-component at Maitri. With the onset of disturbance the nature of Y-variations changes totally at Maitri. One thus comes to the same conclusion, as from the X-variations discussed earlier, that during Q times Maitri is removed from auroral influence and mainly sees the effect of the Sq current system. During D times it is clearly within the auroral oval and experiences the effect of field-aligned current systems which link the magnetosphere to the ionosphere.

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NOV 1992

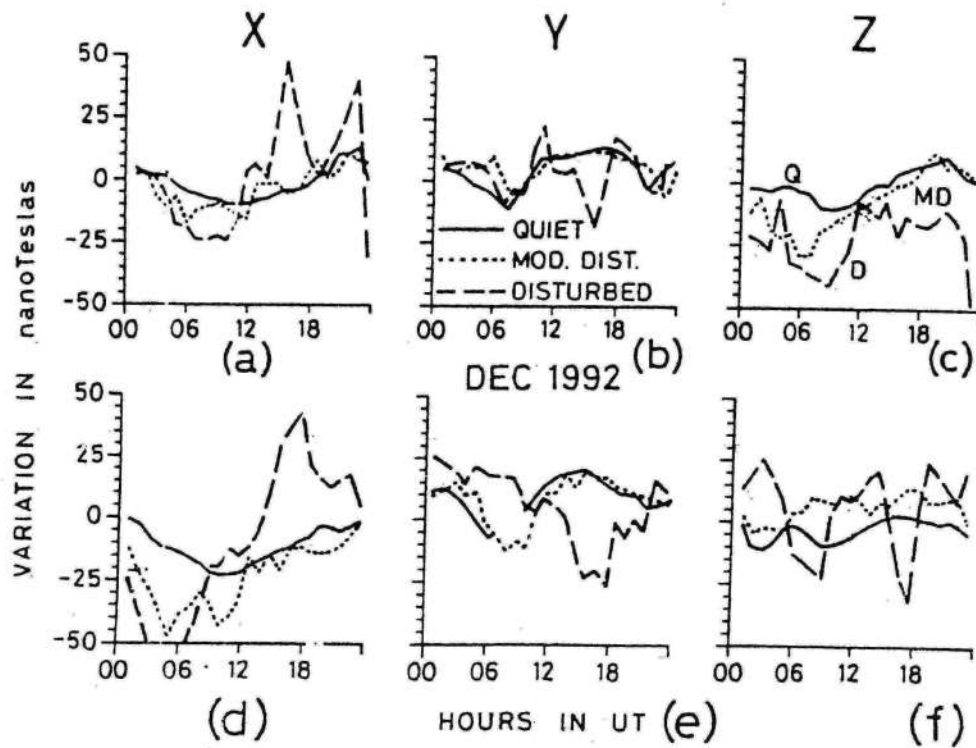


Fig. 7. Hourly variations of X, Y and Z for the Antarctic Summer months November and December 1992

Variations of the Z-component with Magnetic Activity

As a typical example we focus on the Q day variation of the Z- component shown in Fig. 3(f). The variation is practically flat with values in the vicinity of zero. This flat variation of the Z-component is also seen for the Antarctic winter months of May, June, July and August 1992 in Figs. 4(c), 4(f), 5(c) and 5(f) respectively. With the onset of Antarctic summer the Z-curve tends to show a pre-noon negative and a post-noon positive of small magnitude (Figs. 3(c) for March, 6(f) for October, 7(c) for November, and 7(f) for December 1992 respectively). April 1992 in Fig. 3(f) and September 1992 in Fig. 6(c) show a transition behaviour.

With the onset of disturbance the Z-curves (dashed lines) become quite irregular showing marked variations, mainly in the pre-dawn and post-dusk hours. Negative variations characterise the pre-dawn hours and positive variations in the dusk hours. This is seen for almost all the months of April to August 1992 (Figs. 3f, 4c, 4f, 5c, 5f). With the onset of Antarctic summer, however, the pattern becomes more random as shown for October, November, and December 1992 (Figs. 6f, 7c, 7f, respectively).

CURRENT SYSTEMS
WHICH INFLUENCE HIGH LATITUDES
QUIET TIMES

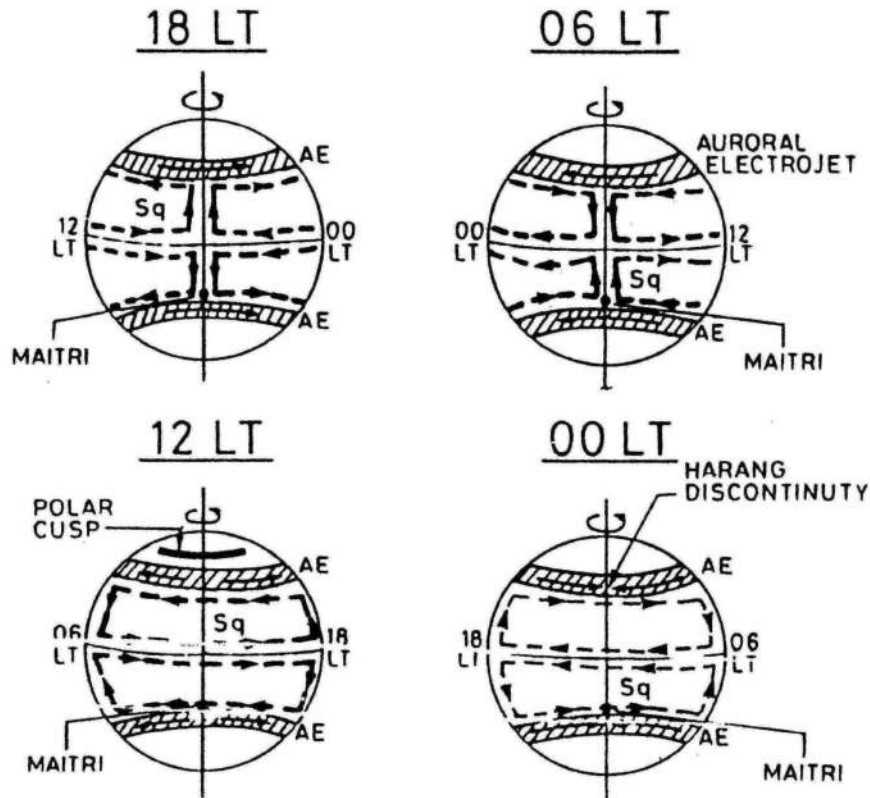


Fig. 8. Directions of Auroral Electrojet (AE) and Solar Quiet time (Sq) current systems during Quiet times

The variations in the Z-component could be the consequence of D time changes in:

- (i) the latitudinal position and intensity of the east-west auroral electrojet currents, and which characterise the midnight sector of the auroral zone, referred to as the Harang Discontinuity (Fig. 8).
- (ii) the north-south directed auroral currents which form the closure currents in the ionosphere for the FAC which link the magnetosphere and the ionosphere.

We are yet to understand how to distinguish between the effects of these two current systems on the Z-variations.

Conclusions

From the foregoing studies it appears that the situation depicted in Figs. 1(a) and 1(b) are correct. During Q conditions, the variations of the X-

(north-south) component suggest that Maitri lies equatorward of the auroral zone and experiences the geomagnetic effects of the ionospheric Sq current system which prevails at low and mid-latitudes. In contrast, during D conditions the auroral oval expands and moves equatorward and Maitri comes directly under the auroral electrojet (AE) current system.

The same is borne out by the variations of the Y-component. During Q conditions, the Y- (east-west) component shows the influence of the Sq current system, but during D conditions, the influence is that of the field-aligned current (FAC) systems which link the magnetosphere to the ionosphere over Maitri. These AE at auroral latitude and the Sq currents at low-mid latitudes are shown in Fig. 8.

As far as the Z-component is concerned its variations during D conditions indicate the passage of Maitri through the midnight sector of the auroral zone called the Harang Discontinuity (Happner, 1972). Fig. 1 of Rostoker (1987) clearly suggests that if Maitri occupied a position on the equatorward side of the auroral oval, the ground magnetic signature of Z would be a positive one on the duskside (due to the EAE) crossing over to a negative one of the dawnside (due to the WAE), as the Earth is rotating. This is what is seen in our study, and is yet another support to our belief that Maitri clearly lies in the auroral oval during magnetically disturbed conditions.

In a following study of this type, using Quiet and Disturbed variations of the X- and Y-components, the intensities of the horizontal current densities of the ionospheric auroral currents over Maitri are being estimated and will be presented elsewhere shortly.

Acknowledgements

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